lamTask 04: Automotive Domain Research & Process Mapping

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Item 1: Five Automotive AI Agent Use Cases

The common pattern across all five: each replaces a manual process where a person stitches together scattered data. The agent retrieves, reasons, and explains in plain language, while the human still makes the decision.

1. Recall Lookup

Take a VIN, query the manufacturer system and NHTSA, cross-reference to find what is genuinely open versus already fixed, and explain it plainly. Value: faster service, fewer missed safety recalls, less liability.

2. Predictive Maintenance Alerts

Watch fleet sensor data (engine temp, vibration, brake wear) and predict failures before they happen. Produces an alert and recommendation, not a repair. Value is timing: converts an expensive roadside breakdown into a cheap scheduled appointment.

3. Warranty Claim Triage

Sort thousands of daily reimbursement claims into buckets: clearly covered (fast track), clearly not (flag), ambiguous or suspicious (escalate). The hard part is spotting claims where the symptom does not match the repair. Reads unstructured technician notes.

4. Parts Catalogue Search

A mechanic describes a part in plain English; the catalogue lists it under a cryptic part number with no shared words. This is semantic search, the same architecture as a RAG pipeline, pointed at a parts database. Filters by vehicle and explains its pick.

5. Dealer Communication Automation

Answer inbound customer messages using access to live dealership systems, so it gives a real answer rather than a canned reply. Handles the routine majority and escalates the rest, such as an angry customer or a warranty dispute.

Item 2: Key Data Sources and Standards

NHTSA Recall Data

NHTSA is the US National Highway Traffic Safety Administration. It maintains the official public database of vehicle safety recalls and offers an API you can query by VIN. This is the authoritative federal source a recall agent checks against, alongside the manufacturer's own system.

VIN Decoding

A VIN is the 17-character Vehicle Identification Number unique to every car built since 1981. The characters are not random; each position encodes specific information: characters 1 to 3 the manufacturer and country, 4 to 8 the vehicle features and engine, 9 a check digit, 10 the model year, 11 the plant, and 12 to 17 the unique serial number. Decoding means parsing those positions to identify exactly what the vehicle is, which is required before checking recalls or matching parts. NHTSA offers a free VIN decoder API.

OBD-II / Telematics

OBD-II is the standardized onboard diagnostics port in modern cars, exposing engine data and fault codes. Telematics is the practice of streaming that data off the vehicle for remote monitoring. Together they are the data source behind predictive maintenance.

FMVSS / NHTSA Reporting Requirements

FMVSS is the set of Federal Motor Vehicle Safety Standards that vehicles must meet. NHTSA reporting requirements are the legal obligations manufacturers have to report defects and issue recalls. These define the compliance and accuracy constraints an automotive agent operates under, where a wrong answer is a safety and legal problem, not just a bug.

Item 3: Team Lead Session Notes

Notes from a session with a project team lead on the automotive agents the team is currently building. The team lead identified cadence automation as the biggest and highest-priority build.

Phase 1 (current)

- MSRP lookup agent: an LLM searches for a vehicle's base price (MSRP, the manufacturer suggested retail price).
- OCR agent (optical image analysis): takes images, reads license plates and VINs, checks for damage, and re-evaluates pricing based on what it finds.

Phase 2 (building)

- Cadence automation agent (the biggest one): automates dealership-to-seller communication. Automated outreach to sellers, scheduling based on availability, and voice, calling, and email handling. Also takes transcripts of live and previous seller conversations and gives advice to the business development staff.
- Live coach agent: reviews daily tasks and gives suggestions across the board, for example flagging missed tasks or tasks that are falling behind.

To confirm

My current understanding is that the dealership is acquiring vehicles from sellers. Worth confirming this, and the specifics of what data the cadence agent is permitted to store and act on.

Item 4: Recall Check Process Map (Draft)

Background: a recall is when a defect that is unsafe is found on a vehicle, and the manufacturer must fix it for free. On a specific car a recall is OPEN (the fix has not been done yet) or CLOSED (already completed). The recall check answers one question: does this car have any open recalls right now?

How a service advisor checks a vehicle for open recalls today, as a manual workflow:

Step 1: Capture the VIN from the customer, windshield, or door jamb.

Step 2: Run the lookup: enter the VIN into the manufacturer system and cross-check NHTSA. Two separate sources that may disagree.

Step 3: Interpret results: separate genuinely open recalls from ones already performed elsewhere. Manual and error-prone.

Step 4: Explain to the customer in plain language what each recall is and whether it is urgent.

Step 5: Schedule the work: check parts and technician availability, then book the repair.

Where time is lost (to confirm)

- Step 2: reconciling two separate systems by hand.
- Step 3: the manual open-versus-completed cross-check, where an open recall can be missed.
- Step 4: translating dense technical recall language into a plain explanation.

Item 5: One-Page Automotive Domain Brief

Focus: Cadence Automation Agent (identified as the team's highest-priority build)

The Problem

A dealership acquiring vehicles has to communicate with a large number of sellers: reaching out, following up, scheduling calls, and tracking each lead across email and phone. Done manually, this is slow and inefficient and prone to human error such as missed follow-ups or leads slipping through the cracks. Every lost or delayed contact is a potential vehicle acquisition the dealership misses.

The Agent's Role

The cadence automation agent takes over seller communication. It handles automated outreach across voice, phone, and email, and schedules interactions based on availability, so follow-ups happen consistently without a person tracking them by hand. It also analyzes transcripts of live and previous seller conversations and surfaces advice to the business development staff. Supporting agents include an MSRP lookup agent for base pricing, an OCR agent that reads plates and VINs from images and assesses damage to re-evaluate pricing, and a live coach agent that reviews daily tasks and flags what is falling behind.

The Data It Touches

Seller contact information, scheduling and availability data, and communication records across email and phone including transcripts of live and past conversations. Supporting agents also touch vehicle images and VINs, pricing and MSRP data, and internal task and workflow data. Much of this is unstructured, such as call transcripts and images, which is why LLM-based agents rather than simple scripts are the right fit.

Compliance and Accuracy Constraints

Because the agents contact real sellers and handle pricing, accuracy and appropriate communication matter. Outreach must be correct and professional, since errors reach real people and affect the dealership's reputation and ability to acquire vehicles. Pricing agents must be accurate, since a wrong valuation has direct financial consequences. More broadly, the automotive domain carries standards and reporting requirements such as NHTSA reporting and FMVSS that any agent touching vehicle or safety data must respect.